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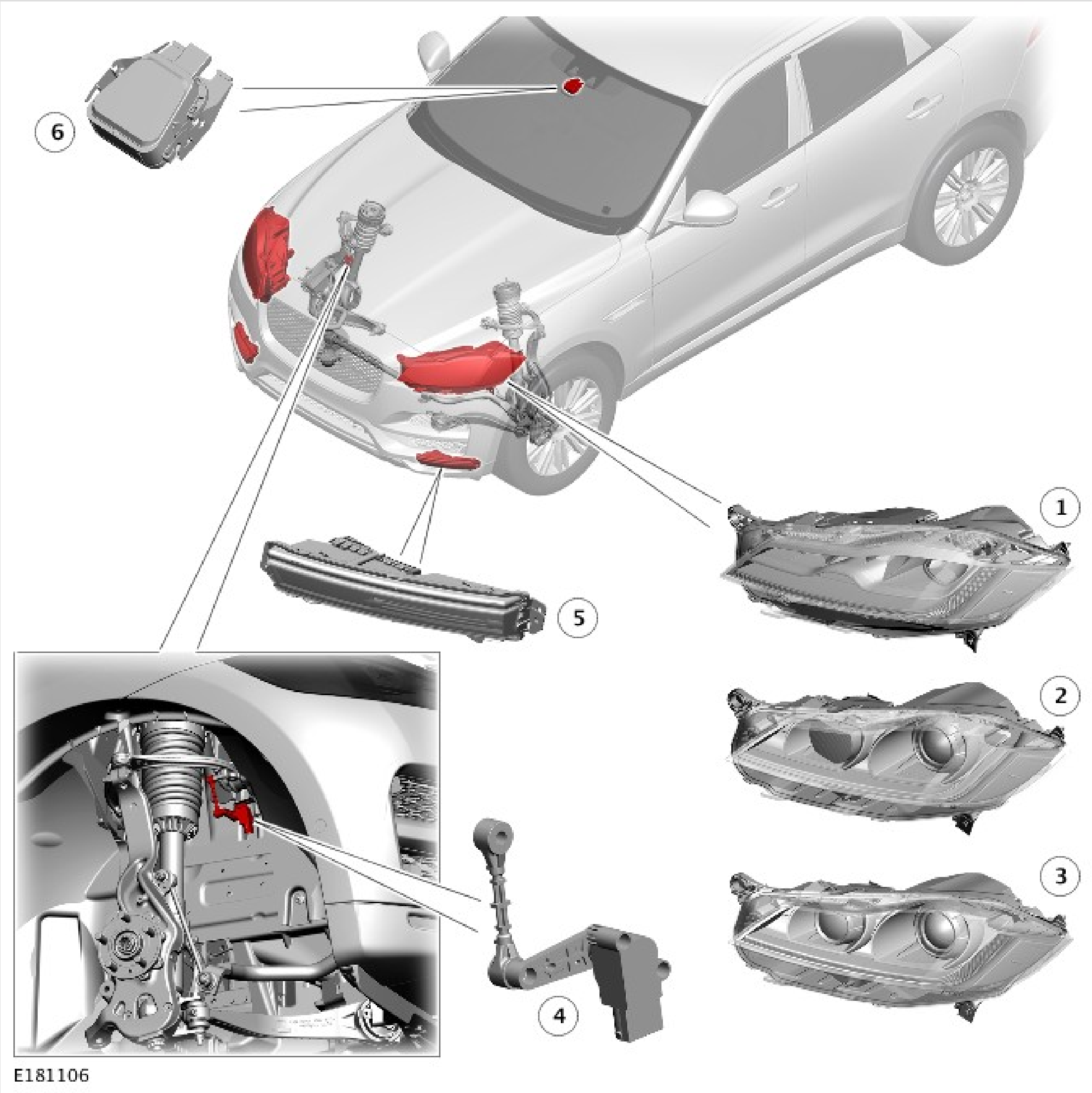
EXTERIOR LIGHTING

DESCRIPTION AND OPERATION

COMPONENT LOCATION 1 OF 3

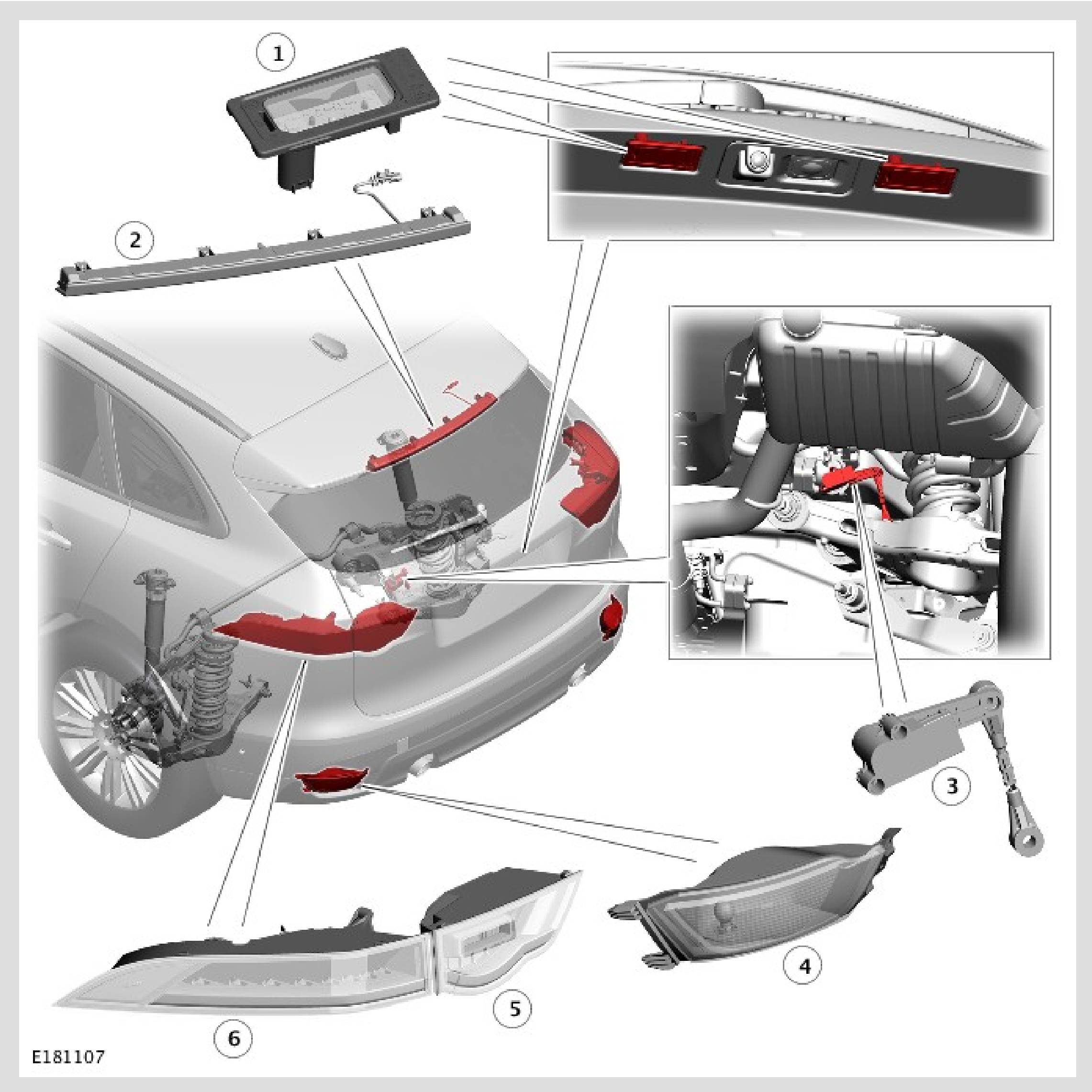


COMPONENT LOCATION - 1 OF 3

ITEM	DESCRIPTION
1	Light Emitting Diode (LED) headlamp - 2 off
2	Halogen headlight (2 off)
3	Xenon headlamp (2 off)
4	Front right height sensor (Xenon and Light Emitting Diode (LED) headlamps only)

5	Front fog lamp (2 off)
6	Rain/light sensor.

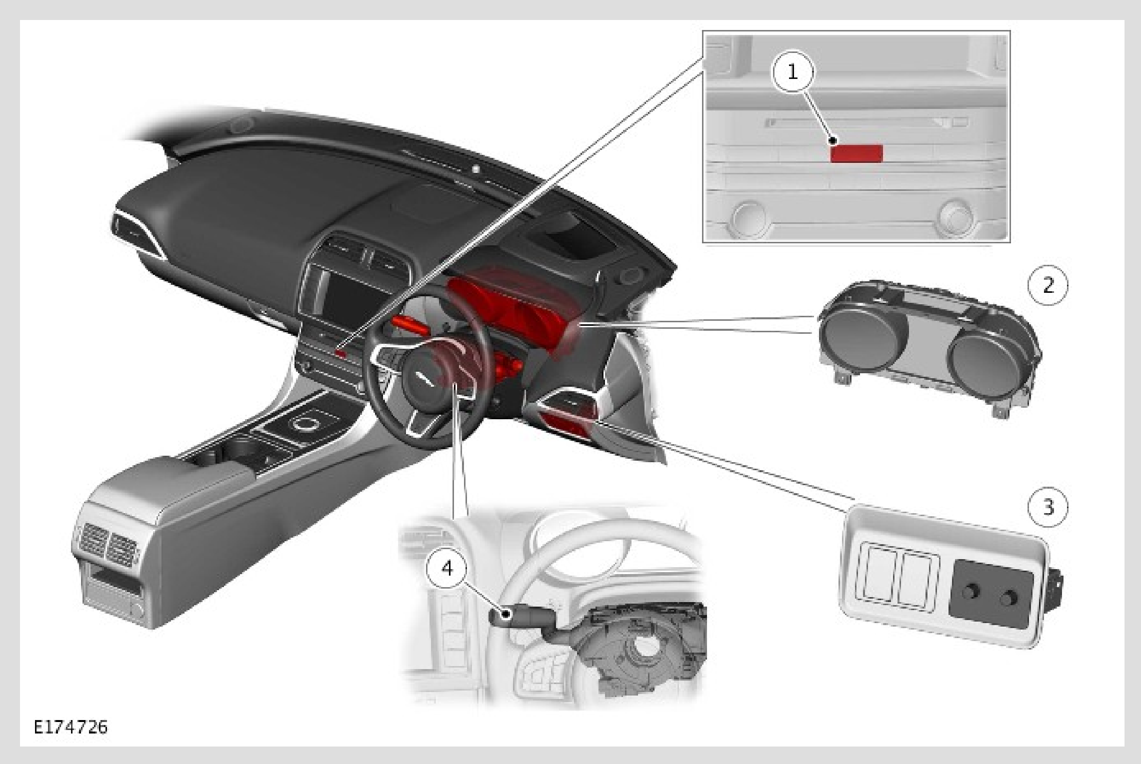
Component Location 2 of 3



COMPONENT LOCATION - 2 OF 3

ITEM	DESCRIPTION
1	License plate lamps (2 off)
2	High Mounted Stop Lamp (HMSL)
3	Rear height sensor (Xenon and Light Emitting Diode (LED) headlamps only)
4	Rear fog lamp assembly (2 off)
5	Tail lamp assembly - inner (2 off)
6	Tail lamp assembly - outer (2 off)

COMPONENT LOCATION - 3 of 3



COMPONENT LOCATION - 3 OF 3

ITEM	DESCRIPTION
1	Hazard warning lamp switch
2	Instrument Cluster (IC)
3	Auxiliary lighting switch
4	Left steering column multifunction switch

OVERVIEW

EXTERIOR LIGHTING

The exterior lighting consists of:

1. Two headlight assemblies, each containing a:

- Headlamp
- Static bending lamp (Not in NAS market)
- Turn signal indicator lamp
- Combined Daytime Running Lamp (DRL) and side lamp
- Side marker lamp and reflector (NAS only).

2. Two tail lamp assemblies, each containing a:

- Turn signal indicator lamp
- Tail lamp assembly
- Stop lamp
- Rear side markers
- Reverse lamps

3. A lighting control switch and steering column multifunction switch
4. Body Control Module/GateWay Module (BCM/GWM)
5. A High Mounted Stop Lamp (HMSL)
6. Front and rear height sensors (Xenon and Adaptive Front lighting System (AFS) Light Emitting Diode (LED) headlamps)
7. Two license plate lamps
8. Auxiliary lighting switch
9. Reverse lamp switch
10. Stop lamp switch
11. An Image Processing Module (IPM) (model/market dependant)
12. A hazard warning lamp switch
13. Warning indicators
14. Rain/Light sensor
15. Battery Junction Box (BJB)
16. Front and rear fog lamps.

The exterior lighting is controlled by the Body Control Module/GateWay Module Assembly (BCM/GWM) as follows:

- Control and monitoring of the exterior lamps including the turn signal indicators and hazard warning lamp functionality.
- Monitoring and evaluation of control inputs from other system control modules and output of applicable messages in the instrument cluster message center.

The BCM/GWM is connected to the High Speed (HS) Controller Area Network (CAN) buses. The BCM /GWM contains a microprocessor which performs the control, monitoring and evaluation of functions.

Driver lighting selections are made using the lighting control switch and left steering column multifunction switch, the rear fog lamp switch and the hazard warning lamp switch. The BCM/GWM operates the stop lamps using the inputs from the Stop lamp switch.

Depending on model and market specification, the lighting system may have:

- An Auto High Beam (AHB) function, where the headlamps are automatically switched between low and high beam in response to signals from the IPM.
- An Autolamps function, where the exterior lights are automatically turned on or off in response to signals from the rain/light sensor used for automatic wiper operation. For additional information, refer to: Wipers and Washers (501-16 Wipers and Washers, Description and Operation).

Driver lighting selections using the left steering column multifunction switch are passed to the BCM /GWM via the clockspring on a Local Interconnect Network (LIN) bus. The BCM/GWM provides circuit protection for all exterior lighting circuits.

The lighting system has an 'auto' lights function which is controlled by the BCM/GWM on receipt of signals from the rain/light sensor located at the top of the windshield. The exterior lights are turned on or off in response to ambient light signals from the rain/light sensor on a LIN bus connection to the BCM/GWM . The auto lights can also be activated when the windshield wipers are activated by signals from the rain sensor, or when the driver activates the wipers in the continuous wipe position for more than 20 seconds.

Three levels of headlamp specification are available:

- Halogen
- Xenon
- AFS LED.

In certain markets the headlamps feature a static bending lamp which illuminates the area at the side of the vehicle when turning into driveways for example.

The tail lamps comprise of an inner and an outer tail lamp assembly.

The tail lamp assembly - outer comprises:

- A turn signal indicator
- A tail lamp
- A stop lamp
- A side marker lamp.

The tail lamp assembly - inner comprises:

- A stop lamp
- A reverse lamp

The rear lamp assemblies require removal for bulb replacement.

All versions of the headlamps have impact resistant polycarbonate lenses. Removable covers at the rear of the headlamps allow for bulb replacement. The headlamps require removal for bulb replacement.

The headlamps use a projector unit. The xenon headlamps use a D3S xenon bulb which operates in both low and high beam. Halogen projector headlamps use a HB3 halogen bulb which also operates in both low and high beam.

Headlamp washer jets are a standard fitment on xenon headlamps.For additional information, refer to: Wipers and Washers (501-16 Wipers and Washers, Description and Operation).

Static dynamic leveling is standard on xenon and AFS LED headlamps (Not in NAS market). The static dynamic system uses height sensors fitted to the front and rear suspension which process the height sensor signals and adjust the headlamp vertical alignment accordingly.

Manual headlamp leveling is used only on vehicles fitted with halogen projector lamps. A rotary leveling thumbwheel is located in the auxiliary lighting switch in the instrument panel, adjacent to the steering column.

Turn signal indicators and high and low beam functions are controlled from the left steering column multifunction switch. The turn signal indicators have a lane change feature. A single operation of the multifunction switch in either direction will operate the selected turn signal indicators for 3 cycles.

An Image Processing Module (IPM) system may also be fitted as an optional feature which automatically controls the high beam headlamps.

DESCRIPTION

HEADLAMP ASSEMBLIES

Three headlamp variants are available depending on model specification:

- Halogen
- Xenon
- Adaptive Front lighting System (AFS) Light Emitting Diode (LED).

The headlamps are sealed units, with scratch resistant polycarbonate lenses bonded to the headlamp body. There are two types of headlamp available, a high line and a mid/low line. the high line has no access as it is a sealed unit. the mid/low line has one sealed access cover and a sealed bulb holder to provide a watertight environment for the headlamp internal components. To prevent fogging of the lens and to allow the headlamp unit to 'breath' in response to internal temperature changes, a vent is located at the outer rear face of the headlamp body. The vent is covered by a Gortex waterproof membrane. This allows ventilation of the headlamp while preventing the ingress of water.

The high line headlamp has no access covers. The mid/low line has an access cover for the Daytime Running Lamp (DRL)/turn signal indicator lamp and a second to access the Xenon or Halogen lamp. Access to the DRL is available via the open hood. Access to the Halogen or Xenon lamp is via the wheel arch, the removable cover in the wheel arch will have to be removed. The lamp cover is removed by rotating clockwise to allow access to the xenon or halogen bulb.

On NAS vehicles, the side marker lamp Light Emitting Diode (LED) is colored amber. The side marker lamp lens is designed so that light from the LED also illuminates the amber colored side marker reflector area at the side of the lamp without the need for an additional bulb.

Headlamp Leveling

Headlamp leveling provides for the adjustment of the vertical aim of the headlamps to minimize glare to other road users when the vehicle attitude changes due to vehicle loading.

Two types of headlamp leveling are available dependent on the type of headlamps fitted to the vehicle:

- Manual headlamp leveling - Halogen headlamps only
- Automatic headlamp leveling - Xenon and Adaptive Front lighting System (AFS) Light Emitting Diode (LED) headlamps only.

Manual Headlamp Leveling - Halogen Headlamps Only

The manual system comprises the following components:

> **NOTE:**
>
> Headlamp leveling is not available on NAS vehicles.

- Two headlamp leveling motors
- Headlamp leveling rheostat rotary control.

A rotary thumbwheel controller is located adjacent to the instrument panel dimmer control in the auxiliary lighting switch. The rotary thumbwheel controller is connected to a rheostat which gives a variable output to the headlamp leveling stepper motors. The motors respond to the output and move to adjust the headlamp vertical alignment as required.

In power mode 6 or above the motors in the lamps are driven from the extended ignition relay in the Body Control Module/GateWay Module (BCM/GWM), via a fuse to each lamp motor.

Movement of the leveling rotary control produces a variable voltage output (hardwired connection). The motors react to the supplied voltage and move the headlamp to the requested position which relates to the supplied voltage from the leveling rotary control. The headlamps can be lowered from their un-laden position to compensate for changes in vehicle attitude due to loading.

The control has four defined positions to compensate for a drop in height at the rear of the vehicle and avoid dazzle to oncoming drivers. The positions are defined as follows:

ROTARY CONTROL ROTATION	VEHICLE LOAD
0	Driver only
1	Driver and front seat passenger
2	Driver and passengers in all seats
3	Maximum gross vehicle weight or maximum rear axle load

Automatic Headlamp Leveling - Xenon and Adaptive Front lighting System Light Emitting Diode Headlamps Only

Automatic headlamp leveling is only available on vehicles with xenon and Adaptive Front lighting System (AFS) Light Emitting Diode (LED) headlamps. The system is not a dynamic headlamp leveling system and changes in vehicle inclination due to positive and negative acceleration are not compensated.

Automatic headlamp leveling provides for the static, periodic adjustment of the vertical aim of the headlamps to minimize glare to other road users when the vehicle attitude changes due to loading.

Automatic headlamp leveling is controlled by the Body Control Module/GateWay Module (BCM/GWM). The BCM/GWM can process and control the headlamp leveling.

The headlamp leveling system comprises the following components and information from other vehicle systems:

- Front and rear vehicle height sensors
- Two headlamp leveling, vertical adjustment motors
- Ignition in power mode 6 or above.

When the lighting control switch is moved to the side lamp or headlamp position, a Local Interconnect Network (LIN) bus message is passed from the clockspring to the BCM/GWM for the selected function. The BCM/GWM then issues a 'lights on' message on the High Speed (HS) Controller Area Network (CAN) powertrain systems bus.

Signals from the front right and rear right height sensors are used to periodically re-align the vertical aim of the headlamps to their optimum position.

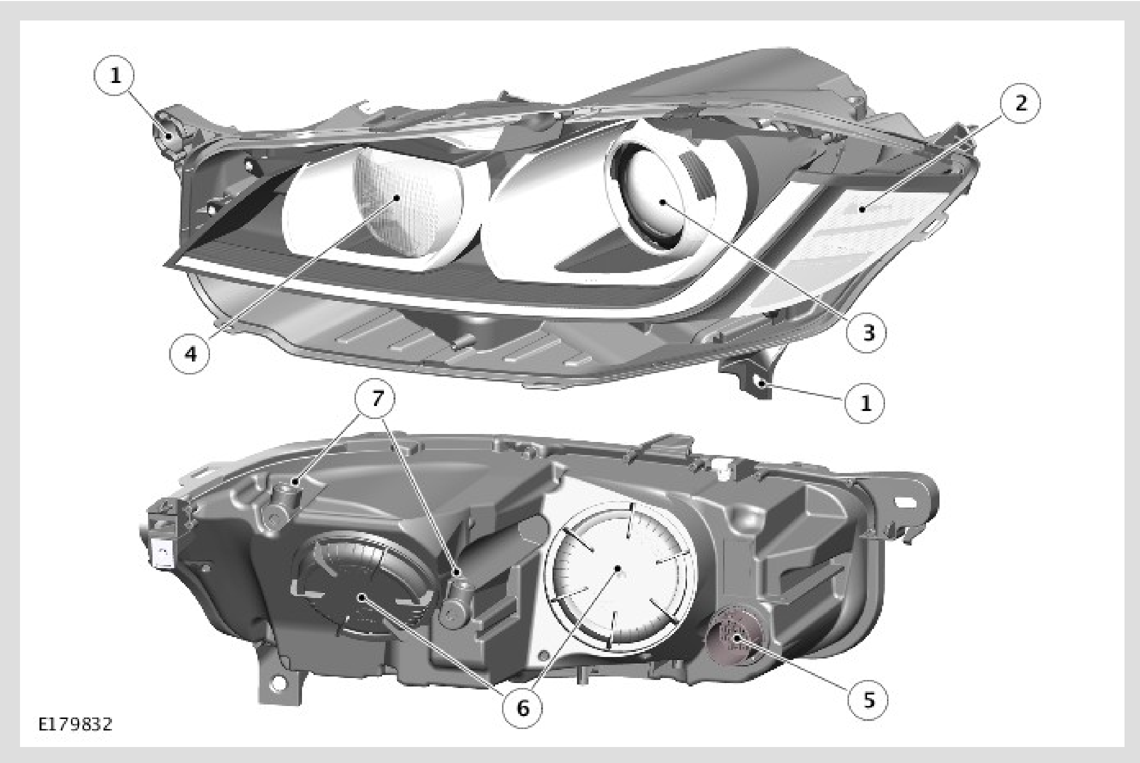
Headlamp Delay

The Body Control Module/GateWay Module (BCM/GWM) controls a headlamp delay function which illuminates after the occupants leave the vehicle. The headlamp delay will operate on low beam headlamps only when the lighting control switch is in the AUTO position and the ignition is off (power mode 0) or in accessory power mode 4.

The headlamp delay is activated when the lighting control switch is in the AUTO position and the engine is switched off. The message centre displays a 'HEADLIGHT DELAY' message and the low beam headlamps will be activated for a period of approximately 30, 60 or 120 seconds. After the delay period, the BCM/GWM automatically switches off the delay function, extinguishing the headlamps. The delay period can be adjusted using the instrument cluster 'Vehicle Settings' menu. The feature can also be disabled using this menu. For additional information, refer to: Information and Message Center (413-08 Information and Message Center, Description and Operation).

The headlamp delay feature can also be switched on when approaching the vehicle or switched off by operating the headlamp switch on the smart key.

HALOGEN HEADLAMP



HALOGEN HEADLAMP

ITEM	DESCRIPTION
1	Headlamp mounting bracket
2	Bezel
3	Projector module
4	Turn signal indicator and Daytime Running Light (DRL)
5	Electrical connector
6	Bulb access cover
7	Adjusters

The halogen headlamp uses two projector modules which comprises ellipsoidal lenses with a solenoid controlled shutter to change the beam output from low to high beam.

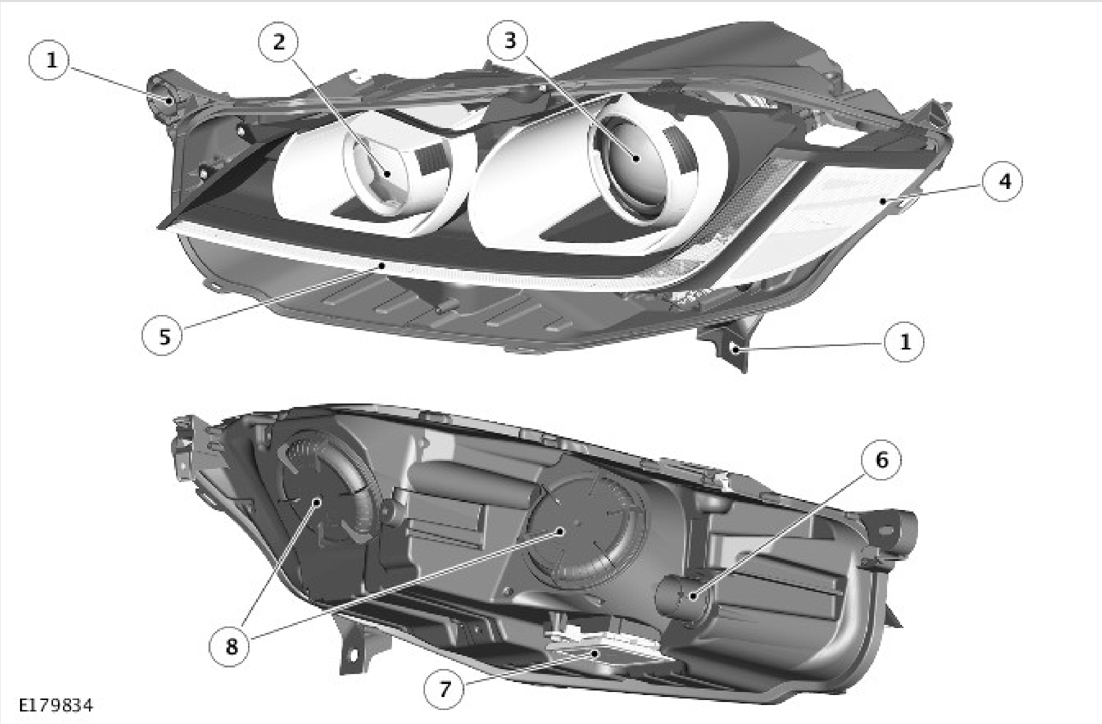
The halogen headlamp is a self contained unit located within the headlamp assembly. The unit comprises a reflector, an adaptor ring, the lens, a shutter controller and the halogen bulb, which together forms an assembly known as the projector module. The reflector is curved and provides the mounting point for the halogen bulb.

The HIR2 bulb is located in an integral holder which locates in a keyway to ensure the correct alignment in the reflector and is secured by rotating to the locked position. The shutter controller is a solenoid which operates the shutter mechanism via a lever. The shutter is used to change the beam projection from low beam to high beam and vice versa.

XENON HEADLAMP

WARNINGS:

- The xenon system generates up to 28000 volts and contact with this voltage could lead to fatality. Make sure that the headlamps are switched off before working on the system.
- The following safety precautions must be followed when working on a xenon headlamp system:
  - DO NOT attempt any procedures on the xenon headlamps when the lights are switched on.
  - Handling of the xenon bulb must be performed using suitable protective equipment, for example gloves and goggles. The glass part of the bulb must not be touched.
  - Xenon bulbs must be disposed of as hazardous waste.
  - Only operate the lamp in a mounted condition in the reflector.



XENON HEADLAMP

ITEM	DESCRIPTION
1	Headlamp mounting bracket
2	Turn signal indicator
3	Projector module - low/high beam headlamp
4	Bezel
5	Daytime Running Lamps (DRL)

6	Electrical connector
7	Xenon Bulb Igniter Module
8	Bulb access covers

The xenon headlamp operates as both a low and high beam unit. The unit comprises a reflector, an adaptor ring, the lens, a shutter controller and the xenon bulb, which together is an assembly known as the projector module. The projector module is installed in a carrier frame that allows the module to tilt for vertical movement of the headlamp beam. Vertical movement is controlled by an actuator, connected between the carrier frame and the projector module, operated by the Body Control Module /GateWay Module (BCM/GWM) for automatic headlamp Leveling.

The reflector is curved and provides the mounting for the xenon bulb. The bulb locates in a keyway to ensure correct alignment in the reflector and is secured by a plastic mounting ring. The bulb is an integral part of the xenon bulb igniter. The shutter controller is a solenoid which operates the shutter mechanism via a lever. The shutter is used to change the beam projection between low and high beam.

The xenon headlamps are controlled by the BCM/GWM using a xenon bulb control module and a xenon bulb igniter for each headlamp. These provide the regulated power supply required to illuminate the xenon bulbs through their start-up phases of operation.

The xenon bulbs illuminate when an arc of electrical current is established between 2 electrodes within the bulb. The xenon gas sealed in the bulb reacts to the electrical excitation and the heat generated by the current flow produces the characteristic blue/white light.

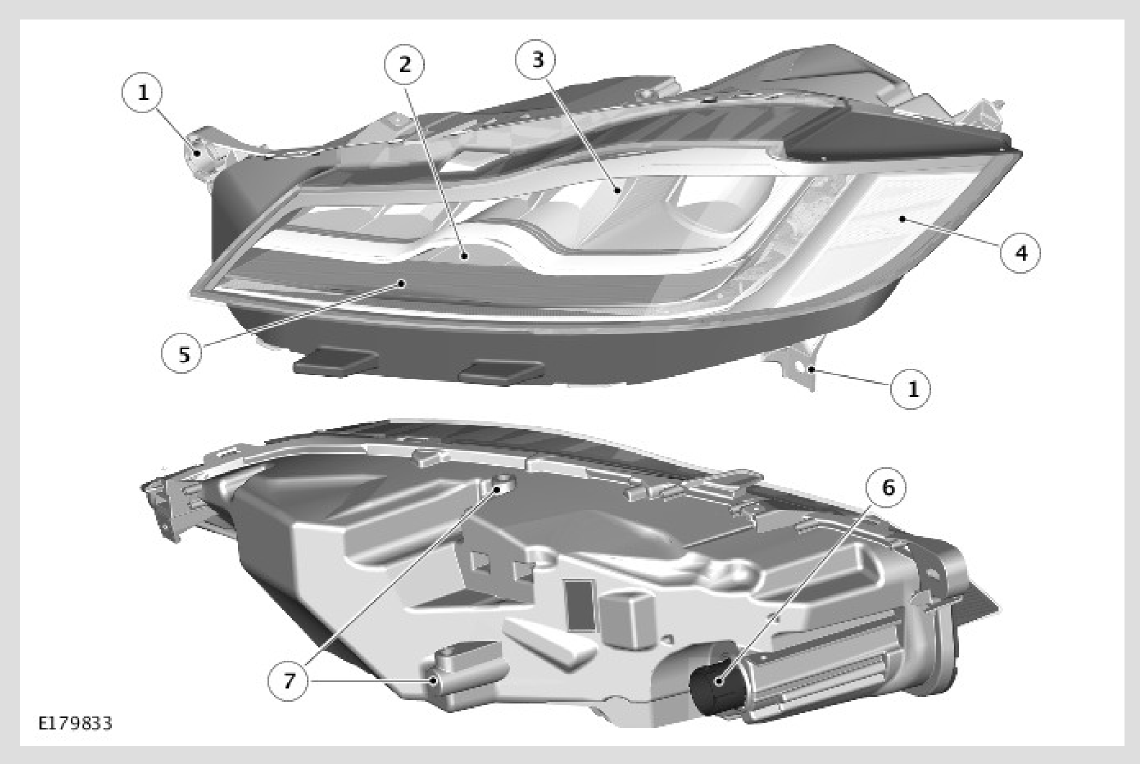
To operate at full efficiency, the xenon bulb goes through 3 stages of operation before the output for continuous operation is achieved. The 3 phases are; start-up phase, warm-up phase and continuous phase.

In the start-up phase, the bulb requires an initial high voltage starting pulse of up to 28000 volts in order to establish the arc. This high voltage starting pulse is produced by the igniter and, once the arc is established, the warm-up phase begins. During the warm-up phase, the xenon bulb control module regulates the supply to the bulb to 2.6A which gives a lamp output of 75W. While in this phase, the xenon gas begins to illuminate brightly and the environment within the bulb stabilizes, ensuring a continual current flow between the electrodes. Once the warm-up phase is complete, the xenon bulb control module changes to continuous phase. The supply voltage to the bulb is reduced and the operating power required for continual operation is reduced to 35W. The process from start-up to continuous phase is completed in a very short time.

The xenon bulb control modules (one per headlamp) receive an operating voltage from the BCM/GWM when the headlamps are switched on. The modules regulate the power supply required through the phases of start-up.

The xenon bulb igniters (one per headlamp) generate the initial high voltage required to establish the arc. The igniters have integral coils which generate the high voltage pulses required for start-up. Once the xenon bulbs are operating, the igniters provide a closed circuit for the regulated power supply from the xenon bulb control modules.

Light Emitting Diode Headlamp



LIGHT EMITTING DIODE HEADLAMP

ITEM	DESCRIPTION
1	Headlamp mounting bracket
2	Light Emitting Diode (LED) High Beam Module
3	Light Emitting Diode (LED) Low Beam Module
4	Bezel
5	Daytime Running Lamp (DRL) and turn signal indicator
6	Electrical connector
7	Adjusters

The Light Emitting Diode (LED) headlamp has a low and high beam unit. Vertical movement is controlled by an actuator, operated by the Body Control Module/GateWay Module (BCM/GWM) for automatic headlamp Leveling.

The LED headlamp is a self contained unit located within the headlamp assembly.

The cooling of the LED control unit allows the warm air to be used for de-icing and demisting the lens of the headlight unit. The cooling of the control unit ensures long life of the LED headlight unit.

The greatest advantage offered by white LEDs lies in the colour of their light, which is also known as the colour temperature. Reaching approximately 6,000° Kelvin the intensity of their light is just about the same as the quality of daylight. When referring to light, Kelvin is a unit of color temperature.

Static Bending Lamps (Adaptive Front lighting System Light Emitting Diode Only)

The Light Emitting Diode (LED) static bending lamps are designed to illuminate the direction of travel when cornering at low speeds. The static bending lamp functionality, which is controlled by the Body Control Module/GateWay Module (BCM/GWM), operates using inputs from the steering angle sensor and vehicle speed information from the Anti-lock Brake System (ABS) control module. The static bending lamp is incorporated into the headlamp assembly. The design of the lens projects a spread of light from the vehicle at approximately 45 degrees to the vehicle axis. The static bending lamp uses LEDs located in the headlamp housing.

NOTE:

The static bending lamps are fitted and operational in Jaguar vehicles, but currently disabled in NAS market vehicles.

The static bending lamps operate with a steering angle sensor CAN bus signal which is received by the BCM/GWM. The BCM/GWM monitors this signal and vehicle speed and activates the static bending lamp LED. When the operation parameters of the lamp are reached, the BCM/GWM snaps the static bending lamp LED on. When the lamp is switched off, the BCM/GWM fades the LED off by decreasing the Pulse Width Modulation (PWM) voltage in a linear manner over a period of 1 second. The static bending lamps can only be active for a maximum of 3 minutes at any given time.

NOTE:

The static bending lamps are fitted and operational in Jaguar vehicles, but currently disabled in NAS market vehicles. It will be enabled when certified.

ADAPTIVE FRONT LIGHTING SYSTEM

The Adaptive Front lighting System (AFS) headlamp assembly contains an additional carrier frame which provides a location for the horizontal actuator. The projector module has a central pivot point which allows the module to move horizontally in response to operation of the horizontal actuator.

The AFS actuators are bi-polar (2 phase) dc stepper motors which are driven by a power output from the AFS power module located in the headlamp assembly. Each stepper motor receives its position information from the Body Control Module/GateWay Module (BCM/GWM) via the applicable AFS power module. When the actuators are powered to their requested positions, a holding current is applied to maintain the actuator position.

The actuators do not supply a positional feedback signal to the BCM/GWM. Each stepper motor requires referencing each time the AFS system becomes active. When the AFS system is active, each vertical actuator is driven to the low beam position and each horizontal actuator is driven to an inboard position until a mechanical stop in the actuator is reached. Once the stop is reached a step counter in the BCM/GWM is set to zero and the actuator is then powered to the operating position as determined by the AFS software.

The AFS is controlled by the BCM/GWM. The BCM/GWM can process and control the headlamp leveling and the AFS headlamp systems.

The system operates by the BCM/GWM receiving inputs from the Powertrain Control Module (PCM) for engine running signal, the Anti-lock Brake System (ABS) control module for steering angle and vehicle speed and a reverse gear input from the transmission.

The BCM/GWM processes these signals and provides an output to the headlamp levelling motors to adjust the headlamp horizontal aim according to vehicle speed and steering angle.

NOTE:

NOTE: In markets with DRL, the AFS system will not operate when the DRL are active.

The BCM/GWM is connected on the High Speed (HS) Controller Area Network (CAN) bus to receive information from other vehicle systems. The BCM/GWM is connected to the AFS headlamps on a dedicated Local Interconnect Network (LIN) bus. The BCM/GWM calculates, using input data from other systems, the required position of the horizontal adjustment of the projector modules. The position information is then output on the LIN bus to the AFS Headlamps. The BCM/GWM outputs the appropriate signals to drive the AFS stepper motors in the headlamp to the appropriate position.

The horizontal position of the projector modules is dependent on a number of input variables. The position is determined by vehicle speed and steering angle. When reverse gear is selected, the projector modules are moved to the straight ahead position to avoid glare to other road users.

The angles of each projector module differ to give the correct spread of light, for example, when turning left, the left hand projector module will have a greater swivelling angle than the right hand projector module.

Adaptive Front Lighting System Operation

When the Body Control Module/GateWay Module (BCM/GWM) receives an ignition on signal, the control module performs the initialization procedure which ensures that the headlamps are correctly aligned on both their vertical and horizontal axes.

The Adaptive Front Lighting System (AFS) swivel initialization starts less than 1 second after the headlamp levelling initialization is activated to ensure that the headlamps are at or below the 0 degree position in the vertical axis, thus preventing glare to oncoming vehicles. The AFS swivel initialization is completed in less than 2.5 seconds. The left and right AFS actuator motors are powered from the 0 degree position to a small movement to the inboard position, then another small movement to the outboard position and then back to the 0 degree position.

Failure Mode

In the event of a failure of the Adaptive Front Lighting System (AFS), a warning indicator in the Instrument Cluster (IC) is illuminated to warn the driver. The AFS warning indicator illuminates when the ignition is in power mode 6 (ignition on) or greater. The AFS warning indicator will also be illuminated if a failure of the steering angle sensor or the vehicle speed signal is detected.

Illumination of the AFS warning indicator does not necessarily mean that there is a fault with the AFS system. The fault may be caused by a failure of another system such as steering angle sensor or the vehicle speed signal missing, preventing the AFS system operating correctly.

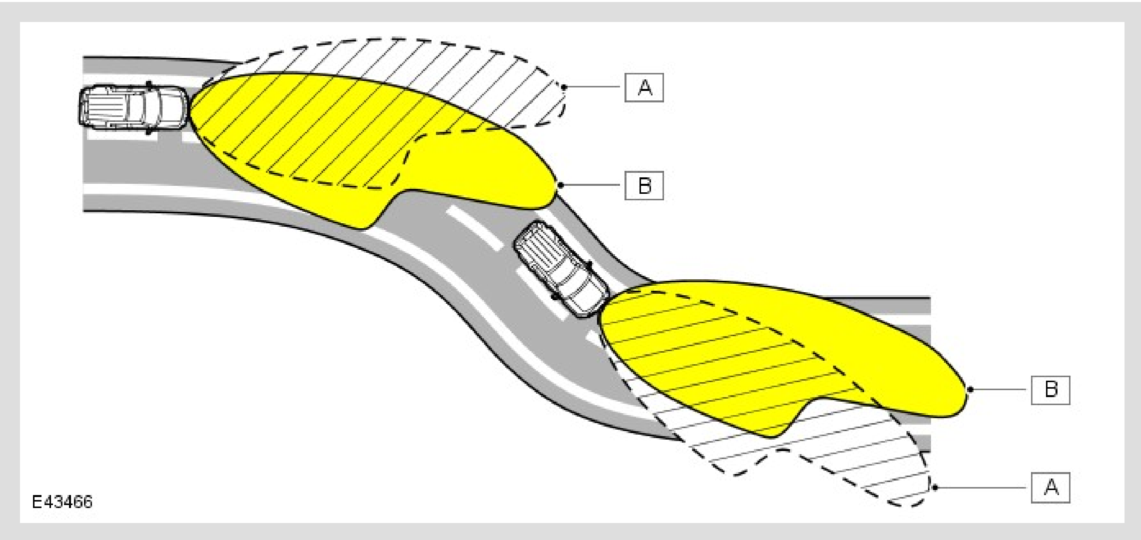
The Body Control Module/GateWay Module (BCM/GWM) performs a diagnostic routine every time AFS is requested. If any fault is found, the BCM/GWM will suspend the operation of the AFS function.

If the AFS levelling system has failed with the xenon projector module in a position other than the correct straight ahead position, the BCM/GWM will attempt to drive the projector module to a position a small amount lower than the standard position.

The BCM/GWM software can detect an internal failure of the BCM/GWM control circuits. The BCM/GWM will power the projector modules to the zero position and prevent further operation.

Faults can be investigated by interrogating the BCM/GWM using a Jaguar approved diagnostic tool to check for fault codes.

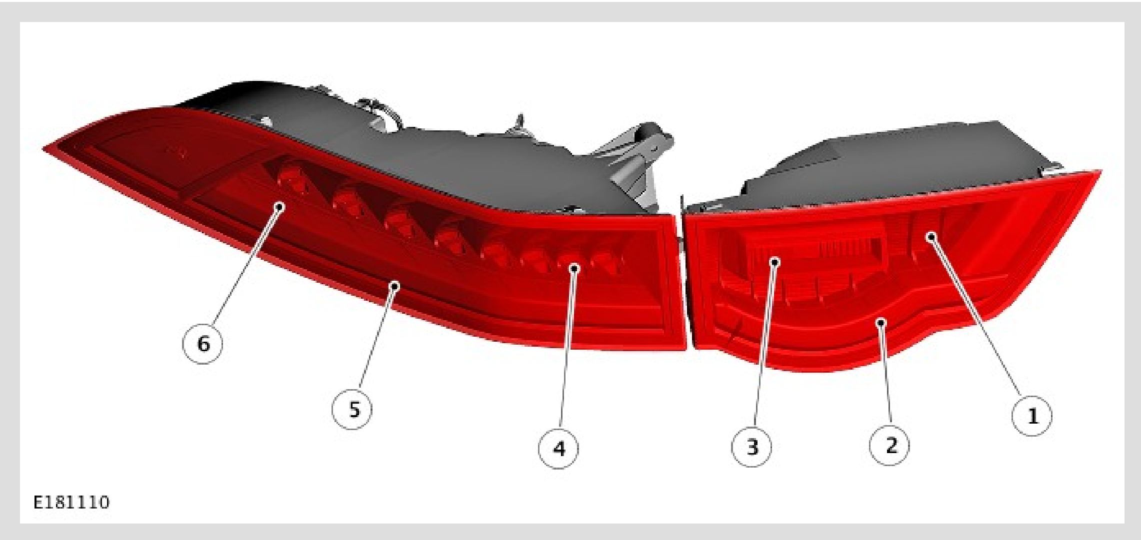
Headlamp Beam Diagram



HEADLAMP BEAM DIAGRAM

ITEM	DESCRIPTION
A	Conventional headlamp beam distribution
B	AFS swivel headlamp beam distribution

TAIL LAMP ASSEMBLIES



TAIL LAMP ASSEMBLIES

ITEM	DESCRIPTION
1	Bezel
2	Inner tail function
3	Reverse lamp
4	Bezel
5	Tail light guide
6	Reflectors

The tail lamp assembly is a two piece unit. The tail lamp assembly - outer is fixed to the body of the vehicle and the tail lamp assembly - inner is fixed to the tailgate.

The tail lamp assemblies have the following functions:

- Tail lamp assembly - outer: Tail/Turn signal indicator and stop lamp
- Tail lamp assembly - inner: Tail and reverse lamp

The tail lamp assembly - outer is located in a recess in the vehicle body and the tail lamp assembly - inner in a recess in the tailgate.

Rear Stop

The stop lamp is an Light Emitting Diode (LED) and is activated when the ignition is in ignition power mode 6 or above and the stop lamp switch is active (by depressing the brake pedal). The High Mounted Stop Lamp (HMSL) will also be activated when the brake pedal is pressed. The stop lamps can also be activated by the Anti-lock Brake System (ABS) when Hill Descent Control (HDC) is active (if fitted). The ABS module sends a High Speed (HS) Controller Area Network (CAN) powertrain systems bus message to the Body Control Module/GateWay Module (BCM/GWM) which supplies power to the stop lamps and HMSL.

Side lamps

The side lamps are operated by selecting side lamps or headlamps on the lighting control switch. The side lamps can be switched on at all times and are not dependent on power mode. The side lamps will also be illuminated when the lighting control switch is in the 'AUTO' position and a 'lights on' signal is received by the Body Control Module/GateWay Module (BCM/GWM) from the rain/light sensor.

Turn Signal Indicator

The turn signal indicator and the tail lamp share the same light guide. The tail light is turned off when the turn signal indicator is operated. The lights are Light Emitting Diodes (LED).

The turn signal indicators are operated by the left steering column multifunction switch or by the hazard warning lamp switch. The left steering column multifunction switch is only active with the ignition in power mode 6 and above, the hazard warning lamp switch is active in all power modes. When active, the turn signal indicators and turn signal indicator in the Instrument Cluster (IC) will flash at a frequency cycle of 400ms on 400ms off.

If a light fails, the remaining turn signal indicator on that side of the vehicle flashes at normal speed. The applicable turn signal indicator in the Instrument Cluster (IC) will flash at double speed with the audible alert at twice the normal rate as well to alert the driver to the failure.

Reverse Lamp

The reverse lamp is located on the tailgate unit and is an Light Emitting Diode (LED).

The reverse lamp is active in power mode 7 and the Body Control Module/GateWay Module (BCM /GWM) receives a reverse selected signal on the High Speed (HS) Controller Area Network (CAN) bus. The automatic and manual transmissions have a reverse switch which senses when reverse gear is selected and engaged and sends the state out on the HS CAN powertrain systems bus to the BCM /GWM.

Side Marker Lamp

The side marker lamp is located in the outer part of the tail lamp assembly, adjacent to rear fog lamp and uses Light Emitting Diode (LED).

The side marker lamp is active at all times when the side lamps are selected on using the lighting control switch. The side marker lamps will also be illuminated when the lighting control switch rotary switch is in the 'AUTO' position and a 'lights on' signal is received by the Body Control Module /GateWay Module (BCM/GWM) from the light sensor.

Fog Lamps

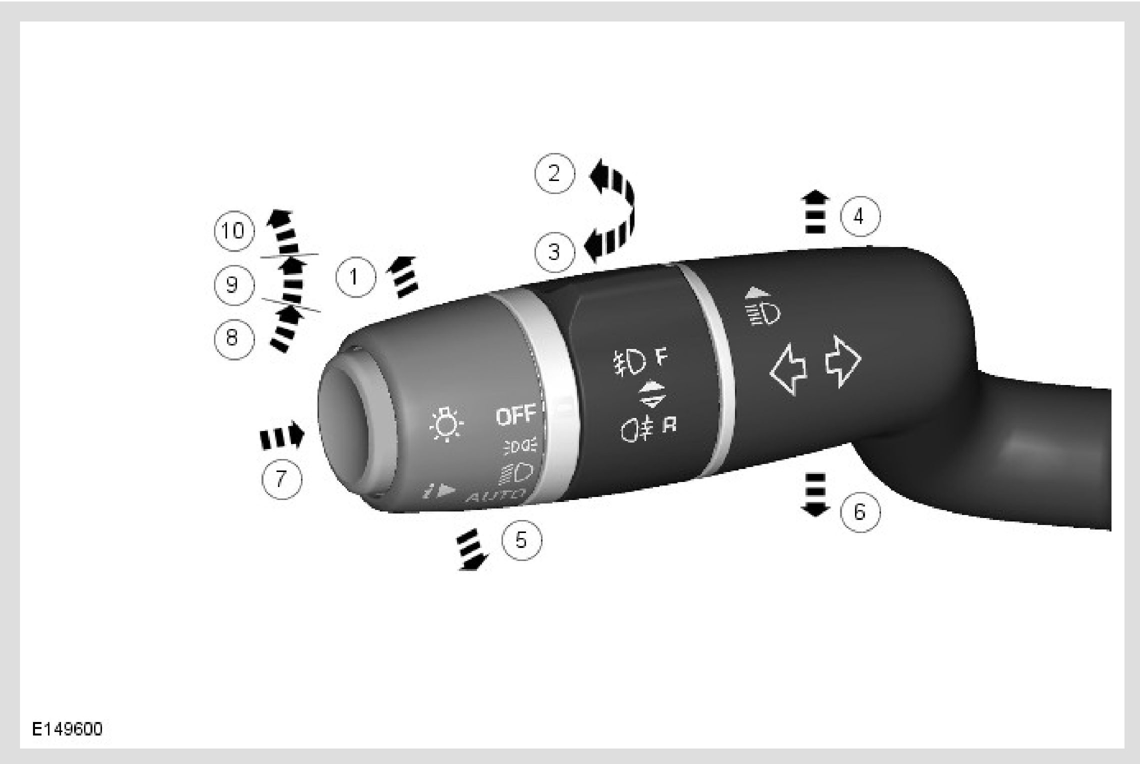
Rear Fog Lamp

The rear fog lamps are controlled by the left steering wheel multifunction switch.

Front Fog Lamp

The front fog lamps are Light Emitting Diodes (LED) and controlled by the left steering wheel multifunction switch.

LEFT STEERING COLUMN MULTIFUNCTION SWITCH



ITEM	DESCRIPTION
1	High beam
2	Front fog lamps
3	Rear fog lamps
4	Right turn signal indicator
5	Headlamp high beam flash
6	Left turn signal indicator
7	Trip computer information
8	Side lamps
9	Low beam
10	Autolamps on

The left steering column multifunction switch allows the following selections:

- All exterior lamps off
- Side lamps on
- Low beam headlamps on
- Autolamps activation
- Headlamp low/high beam
- Headlamp high beam flash
- Left/right turn signal indicators
- Rear fog lamps
- Front fog lamps
- Trip computer functions.

The multifunction switch positions are all connected via a resistive ladder. The output from the resistive ladder is connected to the clockspring which converts the switch operation to Local Interconnect Network (LIN) bus signals. The signals are received by the Body Control Module /GateWay Module (BCM/GWM) which operates the required exterior lighting selection.

Autolamps

The 'AUTO' Autolamps function is a driver assistance system. The driver can override the system operation by selection of side lamp or low beam headlamp on.

The automatic headlamp system uses a rain/light sensor which is connected to the Body Control Module/GateWay Module (BCM/GWM) via a Local Interconnect Network (LIN) bus. The BCM/GWM reacts to the signals from the rain/light sensor and activates the exterior lamps as required.

The light sensor is incorporated in the rain/light sensor located on the inside of the windshield, below the rear view mirror. The wiper system also uses the rain/light sensor for automatic wiper operation. For additional information, refer to: Wipers and Washers (501-16 Wipers and Washers, Description and Operation).

The light sensor measures the ambient light around the vehicle in a vertical direction and also the angular light level from the front of the vehicle. The rain/light sensor uses vehicle speed signals, wiper switch position and the park position of the front wipers to control the system.

The lights may operate in the following circumstances:

- Twilight
- Darkness
- Rain
- Tunnels
- Underground or multistory car parks.

Operation of the autolamps requires the ignition to be on (power mode 6 or above), the left steering column multifunction switch to be in the 'AUTO' position and a lights on request signal from the light sensor. When the 'AUTO' system is active, the side lamp warning indicator in the Instrument Cluster (IC) will be illuminated.

High Beam On and Flash Functions

The high beam is operated by pushing the left steering column multifunction switch towards the instrument panel. The switch will latch in the high beam position. When the high beam headlamps are active, the high beam warning indicator will illuminate in the Instrument Cluster (IC).

The high beam flash function is operated by pulling the left steering column multifunction switch away from the instrument panel. The non-latching switch will operate the high beam headlamps for as long as the switch is held. The switch will return to the high beam off position when released. The high beam warning indicator will illuminate when the high beam headlamps are active.

High beam can also be automatically operated by the Image Processing Module (IPM) system (when fitted), see below.

Turn Signal Indicators

The left and right turn signal indicators are operated by moving the left steering column multifunction switch up or down to select right or left turn signal indicators respectively. The switch will latch in each position.

The switch has a turn signal indicator lane change function which is configurable by the dealer. If the switch is gently pushed, but not latched, to either turn signal indicator position and then released, the applicable turn signal indicators will flash 3 times and then will be automatically cancelled.

If a turn signal indicator fails, the green turn signal warning indicator in the Instrument Cluster (IC) will flash at twice the normal rate and the audible ticking from the IC sounder will also be at twice the normal rate.

Side Lamps and Headlamps

The side lamps and headlamps are selected by a rotary switch on the left steering column multifunction switch.

Rotating the switch from the off position to the side lamps position illuminates the front side lamps, the tail lamps, the license plate lamps and the instrument panel illumination.

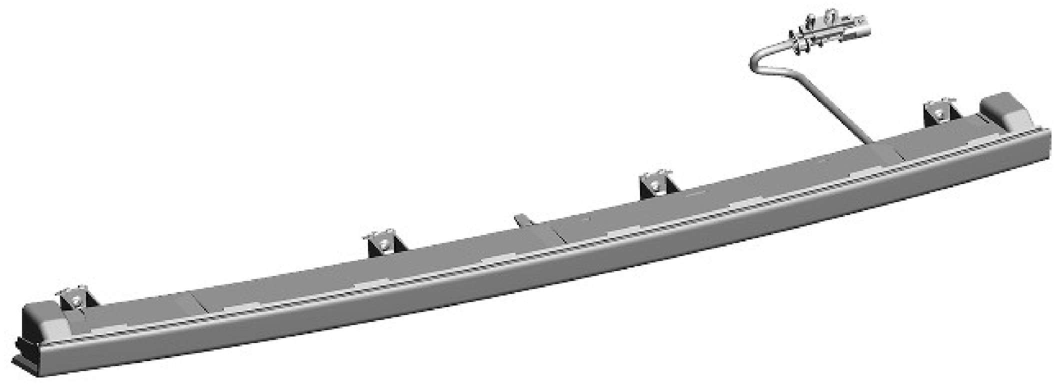
Rotating the switch to the headlamps position, switches on the headlamps in addition to the lamps illuminated by the side lamp position.

Autolamps

Autolamps is selected by rotating a rotary switch on the left steering column multifunction switch to the 'AUTO' position. When the lighting control switch is in the 'AUTO' position, a reference voltage from the Body Control Module/GateWay Module (BCM/GWM) flows through 4 resistors in the lighting control switch. The returned signal voltage is detected by the BCM/GWM which activates the auto headlamp function to activate the headlamps and front and rear side/tail lamps.

The rain/light sensor receives a battery voltage output from the ignition relay in the BCM/GWM. The rain/light sensor continually outputs a Local Interconnect Network (LIN) bus message to the BCM /GWM with information regarding the ambient light levels. When the ambient light level reaches a predetermined value, the BCM/GWM activates the Autolamps feature. The BCM/GWM can also activate the Autolamps when it receives information regarding rain fall from the rain/light sensor which subsequently activates the auto wipers function.

HIGH MOUNTED STOP LAMP



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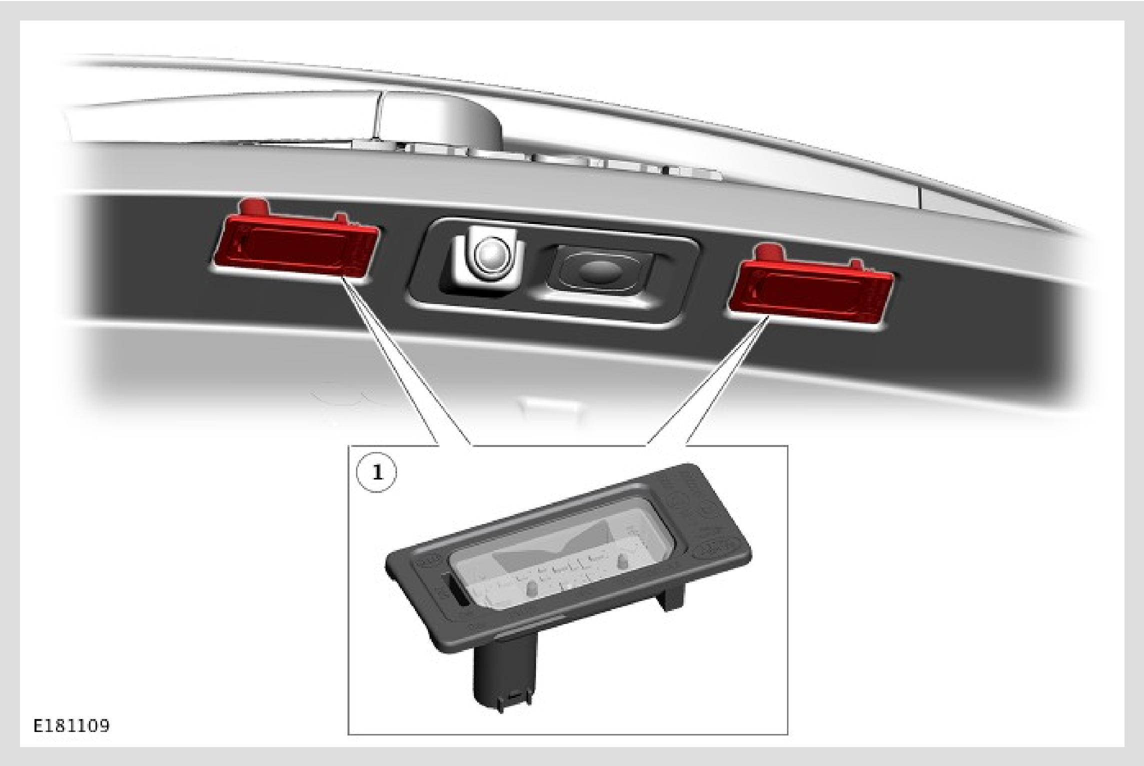
The High Mounted Stop Lamp (HMSL) is located above the rear window. The lamp comprises a plastic housing with a red coloured lens. The lamp is illuminated by Light Emitting Diodes (LED).

The HMSL is activated, along with the tail lamp stop lamps, when the ignition is in power mode 6 or above and the stop lamp switch is active (by pressing the brake pedal).

The HMSL and the stop lamps can also be activated by the Anti-lock Brake System (ABS) when Hill Descent Control (HDC) is active (if fitted). A signal on the High Speed (HS) Controller Area Network (CAN) powertrain bus from the ABS control module is passed to the Body Control Module/GateWay Module (BCM/GWM) which supplies power to the stop lamps.

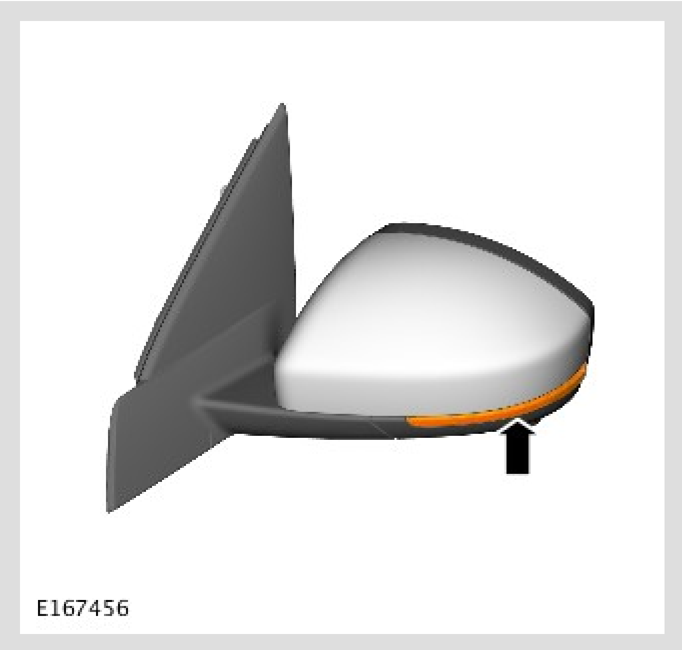
LICENSE PLATE LAMPS



Two license plate lamps are fitted in the tailgate handle, above the license plate in the upper tailgate. Each lamp uses Light Emitting Diodes (LED).

The lamps are secured in the upper tailgate handle with integral clips. The lamps can be released from the handle using a small, flat blade screwdriver. The license plate lamps are active at all times when the side lamps are switched on.

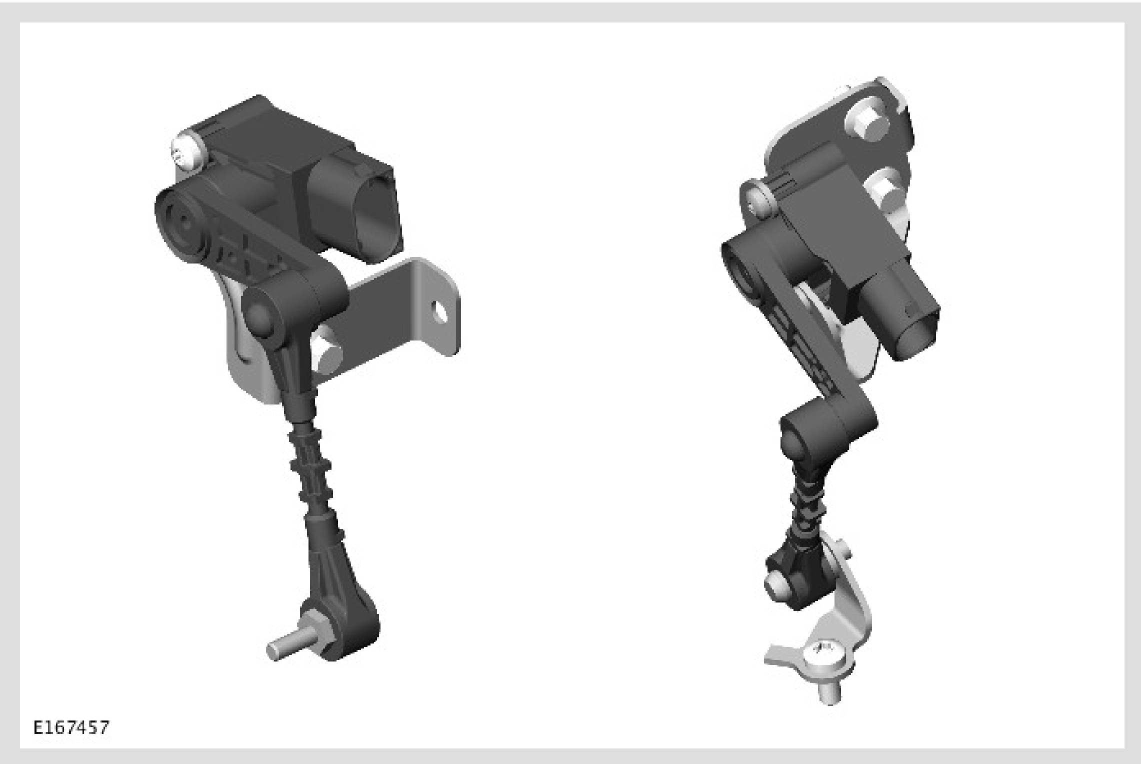
SIDE REPEATER LAMPS



The side repeater lamps are located in the external rear view mirrors.

The side repeater lamps use Light Emitting Diodes (LED). The side repeater lamps have the same functionality as the front and rear turn signal indicator lamps and are operated by the left steering column multifunction switch or by the hazard warning lamp switch. The steering column multifunction switch is only active with power mode 6, the hazard warning lamp switch is active at all times, regardless of the power mode. When active, the side repeater lamps will flash at a frequency cycle of 400ms on and 400ms off. If a side repeater lamp fails, the turn signal indicator lamps continue to flash at the normal rate.

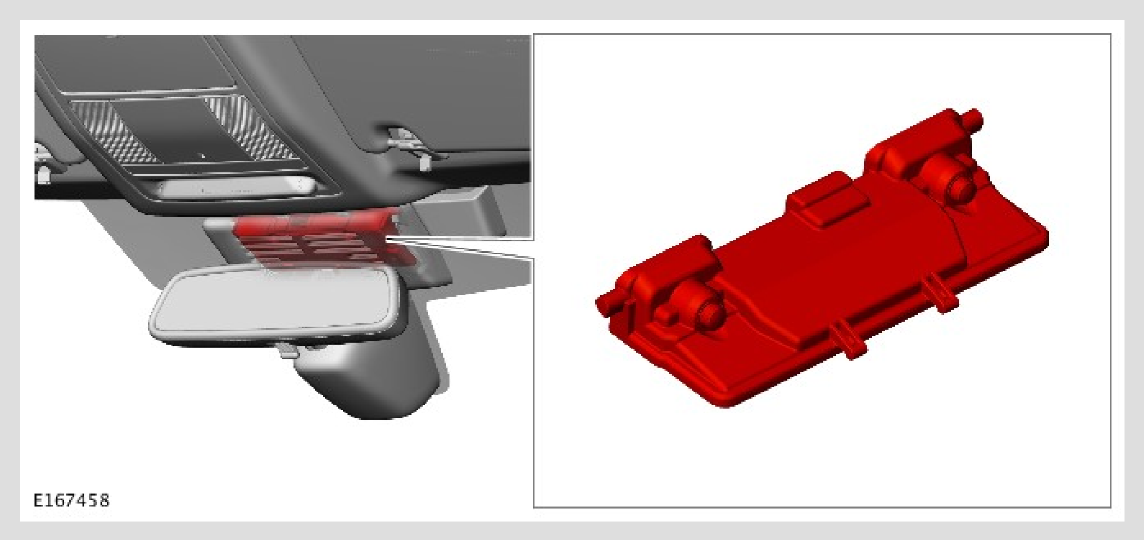
FRONT AND REAR HEIGHT SENSORS



The height sensors used by the headlamp leveling system are both located on the right side of the vehicle.

Each sensor has three connections with the Integrated Suspension Control Module (ISCM); power, ground and signal.

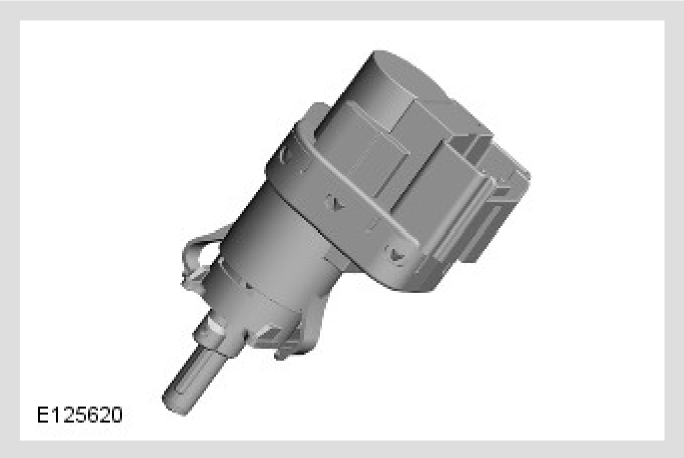
IMAGE PROCESSING MODULE (IPM) CAMERA



Vehicles with the Image Processing Module (IPM) feature have an IPM camera installed in the covers of the rear view mirror support, with the lens directed through the windshield. The camera is connected to the IPM, integrated into the body of the rear view mirror.

The camera is a low resolution image sensor that detects headlamps and tail lamps of the vehicles ahead. The IPM evaluates the image data, checking for light intensity and location. Depending on the image data, the control module then sends a low or high beam request message to the Body Control Module/GateWay Module (BCM/GWM) via the High Speed (HS) Controller Area Network (CAN) chassis bus.

STOP LAMP SWITCH



The stop lamp switch is a two pole switch mounted in the brake pedal bracket and operated by the brake pedal. The output connections of the two poles are hardwired to the Body Control Module /GateWay Module (BCM/GWM). The input connection of one pole is a hardwired signal feed from the BCM/GWM. The input connection of the second pole is hardwired to ground. The BCM/GWM compares the signals from the two poles to confirm the status of the stop lamp switch, which it uses to operate the stop lamps. For additional information, refer to: Anti-Lock Control - Stability Assist (206-09 Anti-Lock Control - Stability Assist, Description and Operation).

HAZARD WARNING LAMPS

The hazard warning lamps are controlled by a non-latching switch in the centre of the instrument panel. The hazard warning lamps operate at all times when selected and are not dependent on the ignition power mode.

When the hazard warning lamps are selected on, all of the turn signal indicators operate as previously described and both left and right turn signal indicators in the Instrument Cluster (IC) also flash. The hazard warning lamps flash at a rate of 400ms on and 400ms off. When the hazard warning lamps are active, they override any request for turn signal indicator operation.

If a trailer is fitted, the trailer turn signal indicators will flash at the same frequency as the vehicle turn signal indicators. The trailer warning indicator in the IC will also flash. If a trailer turn signal indicator bulb is defective, the trailer warning indicator will not flash.

The hazard warning lamps can also be activated by a crash signal from the Restraints Control Module (RCM). This is received by the Body Control Module/GateWay Module (BCM/GWM) which activates the hazard warning lamps. The hazard warning lamps can be cancelled when crash mode is cancelled by the RCM.

TRAILER LIGHTING

Several different types of trailer socket can be fitted to the vehicle depending on market specifications. Refer to the Electrical Reference Library for specific socket details.

The Body Control Module/GateWay Module (BCM/GWM) monitors the turn signal indicators and can detect if more than two lamps are fitted. When a trailer is detected, the trailer warning indicator in the Instrument Cluster (IC) will flash in synchronization with the turn signal indicator.

If one or more of the turn signal indicators on the vehicle or the trailer are defective, the trailer warning indicator will not flash to alert the driver to the bulb failure.

DAYTIME RUNNING LAMPS (DRL)

Daytime Running Lamps (DRL) are a market requirement in certain countries.

For market information and DRL functionality refer to the DRL section. For additional information, refer to: Daytime Running Lamps (417-04 Daytime Running Lamps (DRL), Description and Operation).

Halogen Headlamps

The DRL on halogen headlamps is a dual filament bulb, shared with the side lamp. The DRL operate at a higher intensity than the side lamp illumination.

Xenon Headlamps and Adaptive Front Lighting System Light Emitting Diode Headlamps

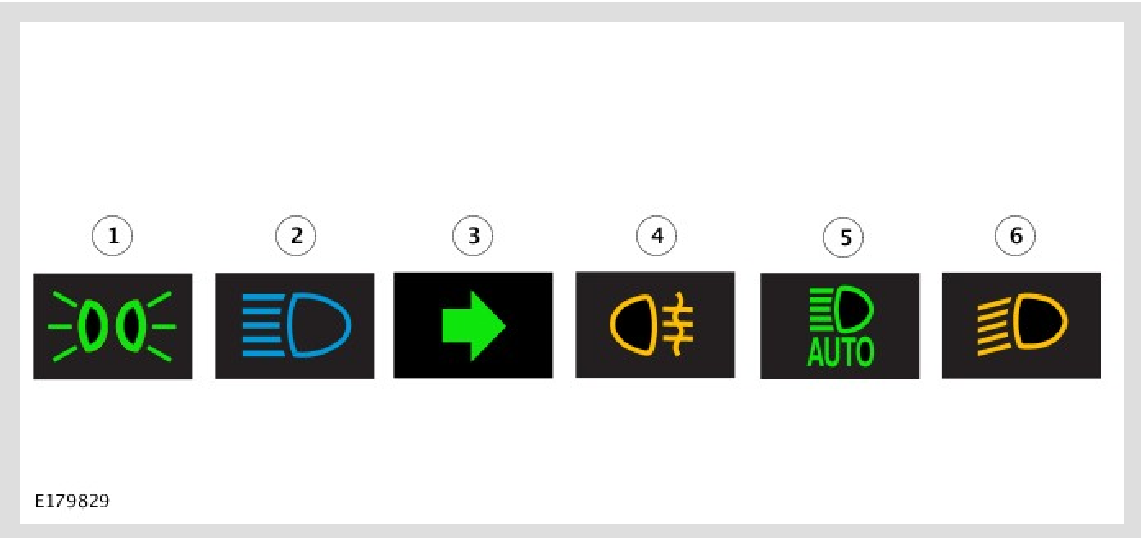
The DRLs and Side lamps use the same string of Light Emitting Diodes (LED). The DRL operate at a higher intensity output than the side lamp illumination.

INSTRUMENT PANEL DIMMER CONTROL

The dimmer rotary thumbwheel controller is located adjacent to the headlamp leveling control (when fitted) in the auxiliary lighting switch. The dimmer control provides a Pulse Width Modulation (PWM) output to control the illumination brightness of the Instrument Cluster (IC), switches and other instrument panel illumination.

The dimmer rotary thumbwheel is connected to a rheostat and a high side switch. The rheostat is a variable resistor which provides a high or low resistance according to its set position. This output is passed to a switchable capacitor or a high side switch. The high side switch uses the output from the rheostat to determine the switching frequency of the capacitor which provides the PWM output of between 8 and 12V to determine the brightness of the illumination.

WARNING INDICATORS



ITEM	DESCRIPTION
1	Side lamps
2	Headlamp high beam
3	Turn signal indicators
4	Rear fog lamps
5	Auto High Beam (AHB)
6	Dynamic Bending & Light Emitting Diode (LED) dipped beam Warning Lamp

The Adaptive Front Lighting System warning indicator is activated by a High Speed (HS) Controller Area Network (CAN) bus message from the Body Control Module/GateWay Module (BCM/GWM) to the

Instrument Cluster (IC). The remainder of the exterior lighting warning indicators are activated by the BCM/GWM using HS CAN powertrain bus messages to the IC.

OPERATION

BODY CONTROL MODULE/GATEWAY MODULE

The Body Control Module/GateWay Module (BCM/GWM) controls the exterior lighting system through interconnections as follow:

- Left steering column multifunction switch
- Side lamp position
- Low beam position
- Automatic ('AUTO') position (if fitted)
- Rear fog lamp switch
- Front fog lamp switch
- Turn signal indicators
- High beam switch
- Headlamp flash
- Stop lamp switch
- Manual headlamp leveling control switch (halogen only)
- Automatic headlamp leveling (Xenon and Adaptive Front Lighting System (AFS) Light Emitting Diode (LED))
- Hazard warning lamp switch
- Rain/light sensor.

Local Interconnect Network

On Xenon and Light Emitting Diode (LED), the Local Interconnect Network (LIN) bus carries a Pulse Width Modulation (PWM) signal. The PWM signal illuminates the position lights (10%) and Daytime Running Lamps (DRL) (100%). The LIN bus also carries feedback from the Static Bending lights, Side Markers, Low and High beam to illuminate the warning indicators in the Instrument Cluster (IC). On LED headlights the LIN bus is used for the levelling and swivel signals.

CIRCUIT PROTECTION

Two 60 Amp fusible links in the BJB protect the power feed to the Body Control Module/GateWay Module (BCM/GWM), left and right lighting circuits respectively. All exterior lighting circuits are protected by Field Effect Transistors (FETs), located in the BCM/GWM, which can detect overloads and short circuits.

The FETs respond to heat generated by increased current flow caused by a short circuit. On a normal circuit this would cause the fuse to blow. The FETs respond to the heat increase and disconnect the supply to the affected circuit. When the fault is rectified or the FET has cooled, the FET will reset and operate the circuit normally. If the fault persists the FET will cycle, disconnecting and reconnecting the power supply. The BCM/GWM stores fault codes which can be retrieved using a Jaguar approved diagnostic system. The fault code will identify that there is a fault on a particular output which assists in fault detection.

ALARM INDICATIONS

The exterior lighting system is used for arm and disarm requests. When the driver locks or unlocks the vehicle, a visual indication of a successful lock or unlock request is displayed to the driver by the hazard flashers. For additional information, refer to: Anti-Theft - Active (419-01 Anti-Theft - Active, Description and Operation).

LIGHTS-ON WARNING CHIME

When the ignition is in the off (power mode 0) or auxiliary (power mode 4) mode and the left steering column multifunction switch is in the side lamp or Low beam position, a warning chime will sound if the driver door is opened. This indicates to the driver that the exterior lights have been left on. The chime is generated from Instrument Cluster (IC) sounder on receipt of a lights-on signal, a door open signal and an ignition off signal from the Body Control Module/GateWay Module (BCM/GWM) on the High Speed (HS) Controller Area Network (CAN) chassis bus.

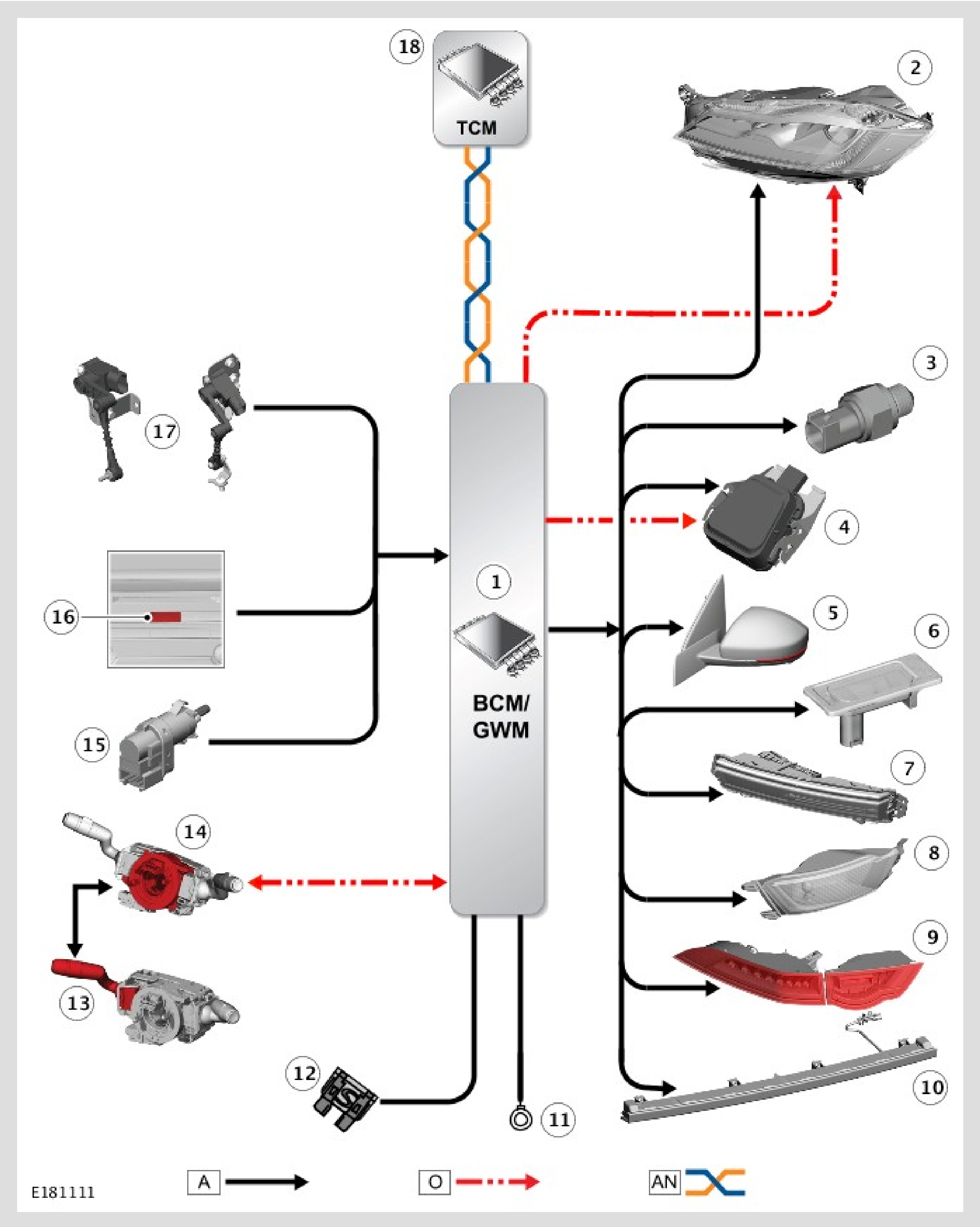
CRASH SIGNAL ACTIVATION

In the event of an accident of a severity to activate and deploy the airbags, the Restraints Control Module (RCM) sends a request to the Body Control Module/GateWay Module (BCM/GWM) via the High Speed (HS) Controller Area Network (CAN) chassis bus to activate the hazard warning lamps. The hazard warning lamps will continue to operate until the ignition mode is changed to the auxiliary power mode 4, or the off power mode 0, or the RCM no longer transmits the crash signal.

HEADLAMP DELAY TIMER

The Body Control Module/GateWay Module (BCM/GWM) controls a headlamp delay function which illuminates the driveway after leaving the vehicle. The BCM/GWM activates the low beam headlamps for the required delay period. The delay timer is set within the Information and Message Center 'Vehicle Set-up Menu'. The default timing is 30s, but the timing can be changed to between 0s (off), 30s (default), 60s, 120s and 240s. For additional information, refer to: Information and Message Center (413-08 Information and Message Center, Description and Operation).

CONTROL DIAGRAM



A = HARDWIRED; O = LOCAL INTERCONNECT NETWORK (LIN) BUS: AN = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) POWERTRAIN BUS.

ITEM	DESCRIPTION
1	Body Control Module/GateWay Module (BCM/GWM)
2	Headlamp assembly (2 off)
3	Reverse lamp switch
4	Rain/light sensor
5	Side repeater lamps in door mirrors (2 off)
6	License plate lamp (2 off)
7	Front Fog Lamp assembly (2 off)
8	Rear Fog Lamp assembly (2 off)
9	Rear Lamps assembly
10	High Mounted Stop Lamp (HMSL)
11	Ground
12	Fuse
13	Left steering column multifunction switch
14	Clockspring
15	Stop lamp switch
16	Hazard warning lamp switch
17	Height sensors (front and rear)
18	Transmission Control Module (TCM)